MIL-W-43105B

28 March 1969
SUPERSEDING
MIL-W-43105A
26 March 1964

MILITARY SPECIFICATION

# WRENCHES, SOCKET, WHEEL STUD NUT, DOUBLE END: AND HANDLES, BAR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope.- This specification covers double end socket wrenches and bar handles for installing and removing wheel stud nuts.

1.2 Classification.- The socket wrenches and bar handles shall be of the following types, sizes and lengths, as specified (see 6.2).

1.2.1 Socket wrenches.-

Type I - Hexagon both ends

Size 1 - For 1-1/2 and 1-1/16 inch stud nuts
Size 2 - For 1-1/2 and 1-3/16 inch stud nuts

Type II - Hexagon one end, square one end

Size 1 - For 1-1/2 inch hex and 13/16 inch square stud nut

Wrench lengths

Length 1 - 14 to 16 inches long
Length 2 - 17 to 19 inches long
Length 3 - 22 to 24 inches long

Size 2 - For 1-3/4 inch hex and 31/32 inch square stud nut.

FSC 5120

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1.2.2 Bars, handle.-

Size A - 3/4 inch diameter round bar; 20 inches long
Size B - 3/4 inch diameter round bar; 30 inches long
Size C - 1 inch diameter round bar; 30 inches long

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-H-15424 - Hand Tools, Packaging of.
MIL-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

ARMY NATICK LABORATORIES

2-9-211 - Wrenches, Socket, Wheel Stud Nut, Double End; Handles, Bar.

(Miniature copy of Drawing 2-9-211, identified as figure 1, is attached for information purposes only).

(Copies of specifications, standard and drawing required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).



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2.2 Other publication.- The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

E 18-65 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), before production is commenced, a sample of the item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The wrenches and the handles furnished under this specification shall be the manufacturer's current standard product except for any changes necessary to comply with specification requirements.

* 3.3 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for socket wrenches, provided the completed item complies with all provisions of this specification.

3.4 Construction.-

3.4.1 Bars, handle.- The handle shall conform to details and dimensions shown on Drawing 2-9-211. Size A and B handles are for use with the type I, size 1 and 2, and type II, size 1 wrenches. Size C handles are for use with type II, size 2 wrenches.

* 3.4.2 Wrenches, socket.- The wrenches shall conform to details and dimensions on Drawing 2-9-211, except that the handle holes may be either round or hexagonal.



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3.4.3 Hardness.- Wrenches and handles shall be heat treated to a Rockwell hardness of not less than RC28 nor more than RC36 when tested in accordance with 4.4.1.

* 3.5 Performance.- The wrenches and handles shall be capable of withstanding the minimum test loads specified in tables I and II without any distortion or permanent deformation when tested as specified in 4.4.2.

TABLE I.- Test loads for sockets

| Nominal Socket sizes (inch) 1/ | | Inch-pounds (minimum) |
|---|---|---|
| Hexagonal | Square | |
| - | 13/16 | 1600 |
| - | 31/32 | 2300 |
| 1-1/16 | - | 2800 |
| 1-3/16 | - | 3800 |
| 1-1/2 | - | 4700 |
| 1-3/4 | - | 7500 |

1/ See Drawing 2-9-211 for exact dimensions.

TABLE II.- Test loads for bars, handle

| Bar handles | Length (inches) | Inch-pounds (minimum) |
|---|---|---|
| Size A | 20 ± 1/2 | 300 |
| Size B | 30 ± 1/2 | 400 |
| Size C | 30 ± 1/2 | 400 |

3.6 Finish.- The wrenches and handles shall have a phosphate coating in accordance with type Z, class 2 of MIL-P-16232.

3.7 Marking for identification.- Wrenches and handles shall be permanently marked with the manufacturers name or readily identifiable trademark.

* 3.8 Workmanship.- Wrenches and handles shall have all sharp edges, burrs, slivers and scale removed.



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## 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects listed in Table III, dimensions specified, and tested in accordance with 4.4.2.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Intermediate test.- Prior to finishing, socket wrenches and handle bars shall be tested for hardness in accordance with 4.4.1. The inspection level shall be S-2 with an acceptable quality level (AQL) of 1.5 defects, expressed in terms of defects per hundred units. The sample unit shall be one wrench or one bar.

4.3.2 End item inspection.- The inspection lot shall consist of all complete wrenches or handles of the same type, style and size, as applicable, offered for inspection at one time. The sample unit for this inspection shall be one wrench or handle bar.

* 4.3.2.1 Visual examination.- The completely assembled wrenches or bars shall be examined for defects in table III. The inspection level shall be II with an AQL of 4.0 expressed in terms of defects per hundred units.



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TABLE III.- Examination of defects

| Examine | Defect |
|---|---|
| Finish (Phosphate coating) | Not as specified |
| Design | Not type, size, or length specified |
| Construction and workmanship | Wrench or bar cracked, bent or otherwise malformed<br>Any sharp edges, burrs, slivers or scale<br>Wrench openings not hexagonal or square as specified<br>Holes for bars missing<br>Holes not round or hexagonal (as applicable) |
| Identification marking | Missing, incomplete, illegible or not permanent |

4.3.2.2 Dimensional examination.- Inspection shall be for compliance with dimensional requirements specified on Drawing 2-9-211. Any deviation from the specific requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

* 4.3.2.3 End item testing.- When a preproduction sample is not required, the first production item shall be tested in accordance with 4.4.2.

* 4.3.2.4 Examination of preparation for delivery.- Examination of preparation for delivery shall be in accordance with MIL-H-15424.

4.4 Tests.-

* 4.4.1 Hardness test.- Socket wrenches and handle bars shall be tested for hardness in accordance with ASTM E 18-65 to determine compliance with 3.4.3.

* 4.4.2 Load test.- Each socket on the wrenches shall be tested. The ends of the wrenches and handles shall be marked before application of the test load for reference. The marking shall be such that any distortion caused by the test shall be visible. The manufacturer shall certify that the test mandrels for each of the sockets have been hardened to a Rockwell C55 minimum. The mandrels shall be furnished to fit securely into each of the sockets.



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4.4.2.1 Wrench test.- The wrench shall have the handle inserted through the wrench hole farthest from the socket being tested. The wrench shall be placed on the test mandrel with one end guided but free to rotate. Force shall be applied to the handle to produce the applicable torque test load as specified in table I.

4.4.2.2 Handle test.- Handles shall be tested by clamping one end in a vise for a distance of 6 inches. The load specified in table II shall be applied to the opposite end of the bar at a distance of 12 inches from the vise jaws for the size A handle and 22 inches for the size B and C handle. Other methods of test may be used, provided they apply a load equal to the product of the force and distance specified herein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging, packing and marking.- Preservation, packaging, packing and marking for level A, B or C shall be in accordance with MIL-H-15424 (see 6.2).

6. NOTES

6.1 Intended use.- The socket wrenches and handle bars covered by this specification are to be used in the removal and replacement of wheel stud nuts on vehicle wheels.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, size and length of socket wrenches and size and length of handle bars required (see 1.2).
(c) First article.- When preproduction sample is required (see 3.1).
(d) Selection of applicable levels of preservation and packaging, and packing (see 5.1).

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, corrections, modifications, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.



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Custodians:

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Review activities:

Air Force - 84, 85

User activities:

Army - AT
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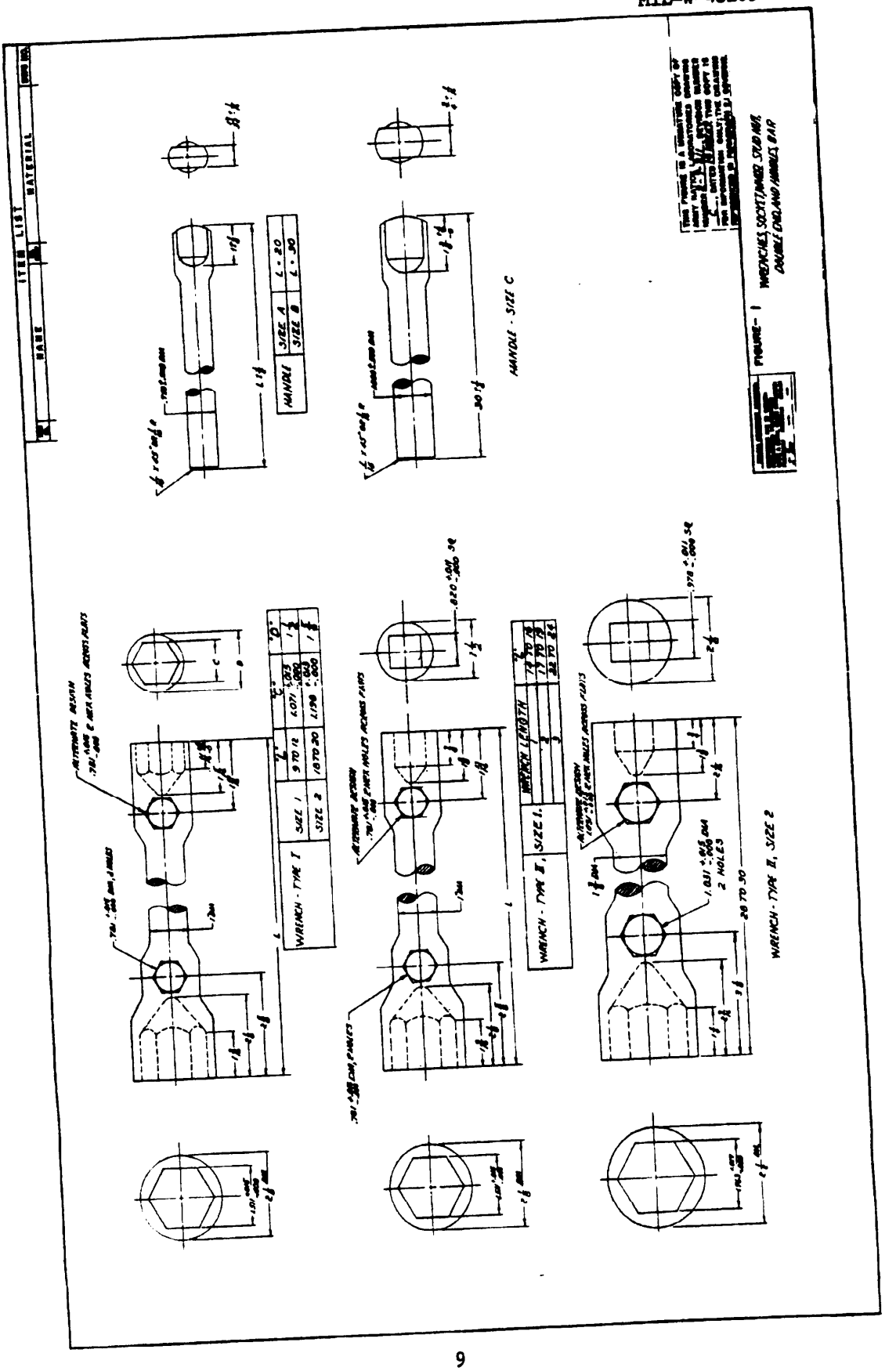
Preparing activity:

Army - GL

Project No. 5120-0572



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

**INSTRUCTIONS:** This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Wrenches, Socket, Wheel Stud Nut, Double End: and Handles, Bar
MIL-W-43105B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 64 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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